# Mathematics assignments

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## 1 Problem 1

Given an automata, how to calculate the (exponential) growth of the language accepted by the automata? In other words, if $c(n)$ is the number of words of length $n$ in the language, how the values $\{\sqrt[n]{c(n)}\}_n$ evolve?

Hint: Each automata can be seen as a directed graph (multi-edges/loops allowed) where we start at a vertex $q_0$ (TODO: check graph-theoretic definitions!). In the case of binary alphabet, we can go to another vertex by following either one of the 2 outgoing edges. Therefore, the number of valid words of length $n$ accepted by an automata is the number of paths from $q_0$ to an accepting state of length $n$ (i.e., $n$ is the number of edges).

Suppose there are $d$ states. We can represent the directed graph as a matrix $M$, where $M_{i,j}$ is the number of edges from $i$ to $j$. It follows that the number of paths from state $i$ to state $j$ of length $n$ is $(M^n)_{i,j}$ (TODO: check the definition of matrix multiplication!). It is not hard to see that the entries of $M^n$ usually grows exponentially. Indeed, denote $\|A\| = \max_{i,j} A_{i,j}$ for any matrix $A$, and follow these steps:

- ($1/2$ hr) Prove that for every $k, \ell$, we have

  $$\|M^{k+\ell}\| \le \text{const}\, \|M^k\| \|M^\ell\|,$$

  where the constant is some value that does *not* depend on the actuall entries of $M$. (Hint: Does it even hold for any $A, B$ instead of $M^k, M^\ell$?)

- (1 hr) A nonnegative sequence $s_n$ is said to be submultiplicative if $s_{n+m} \le s_n s_m$ for every $s_n s_m$. Prove that for such $s_n$, we have

  $$\lim_{n\to\infty} \sqrt[n]{s_n} = \inf_n \sqrt[n]{s_n}.$$

  (Hint: If $\sqrt[m]{s_m}$ is small, then how is $\sqrt[tm]{s_{tm}}$ for $t \ge 1$? Prove the limit using $\limsup, \liminf$ rigorously!)

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- ($^1/_4$ hr) Use the previous point to conclude that

$$\|M^n\|^{\frac{1}{n}}$$

  converges.

`TO BE CONTINUED...`